D2C Customer Profitability & Cohort Unit Economics Intelligence

A full-stack analytics system built to answer the one question every D2C founder avoids: are we actually making money on our customers, or just growing revenue?

8,000 Customers | 12,208 Orders | 36 Months of Data | Model AUC: 0.797 | 5-Page BI Dashboard | 6 SQL Scripts

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THE BUSINESS PROBLEM

A D2C health and wellness brand is growing 40% month on month but burning cash. Leadership assumes the business will turn profitable once customers start coming back. The problem is nobody has checked whether that is actually happening.

This project was built to answer four questions the finance team could not answer from their existing reports:

Which acquisition channels produce customers who pay back their CAC? Are any cohorts profitable at Month 6 or Month 12? What is the true net profit per customer after returns and acquisition cost? Which customers are about to churn and are worth saving?

WHAT WAS BUILT

Dataset: 5 interconnected tables simulated with real D2C business logic. Customers, orders, order items, marketing spend, and returns covering 36 months of activity.

SQL Layer: 6 analysis scripts in PostgreSQL covering acquisition analysis, cohort construction, LTV vs CAC, customer profitability, returns impact, and marketing efficiency.

Python: Full EDA notebook with cohort retention heatmap, RFM clustering using K-Means (K=4), and a Random Forest churn model. Identified and resolved data leakage from the initial model.

Power BI: 5-page interactive dashboard with slicers across channel, city tier, and year. Pages cover executive summary, cohort retention, LTV intelligence, customer profitability, and returns and marketing efficiency.

KEY FINDINGS

LTV:CAC Ratio by Acquisition Channel

Direct 17.07x (Exceptional)
Organic Search 7.74x (Excellent)
Email Referral 6.55x (Excellent)
Paid Google 1.62x (Marginal)
Paid Instagram 1.17x (Barely Viable)
Influencer 0.73x (Destroying Value)

Payback Period by Channel: Direct and Organic customers recover their acquisition cost within a single order. Influencer customers need 3 orders to break even, but most never place a second one.

Churn Model: Random Forest classifier trained on purchase behaviour features. Initial model returned ROC-AUC of 1.0, which was traced to data leakage where recency directly encoded the churn label. After removing the leaking feature, the clean model scored 0.797. Top predictors were purchase frequency and total gross profit.

RECOMMENDATIONS

Stop the bleed on Influencer spend 0.73x LTV:CAC means the channel is losing money on every customer acquired. Shifting 30% of that budget into email retention or content would improve overall blended LTV:CAC meaningfully.

Invest in channels that already work Direct and Organic are running at 17x and 7.7x but are severely underinvested. These customers pay back immediately and stay longer.

Trigger churn intervention at Day 45 Customers with a single order and gross profit under Rs.500 have over 78% churn probability. Waiting until Day 90 to act is too late.

Stop using blended CAC for decisions Blended CAC hides the fact that Influencer CAC is 3x higher than Organic. Every budget decision should use channel-level CAC vs channel-level LTV.

TECH STACK

Python 3.10, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, PostgreSQL, Power BI, DAX, Faker, Statsmodels

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GitHub